MTH203: LECTURE 01

The aim of this lecture is to introduce the central topic of this course – the study of sequences and series of real numbers. The textbook for the course is the following:

> Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015.

In these lecture notes, I will refer to this book as the **textbook**. The syllabus for this course roughly corresponds to the first two chapters of this book. You are also encouraged to other books that are listed in the course description.

1. Sequences: A rough definition

Roughly speaking a sequence of real numbers is something of the following form:

$$a_1, a_2, \ldots, a_n, \ldots$$

Here, for every positive integer i, a_i denotes a real number. The dots indicate that there are infinitely many real numbers in the above collection. The entries may be repeated. Also, it is important to note that this is not just a *collection* of real numbers, the *order* also matters. We will look at a more rigorous definition later.

We will denote the above sequence by $(a_n)_{n=1}^{\infty}$. Note that the notation indicates that n is varying from 1 all the way to "infinity", meaning that it varies through all the positive integers. This may also be written as $(a_n)_{n\in\mathbb{N}}$. Recall that $\mathbb{N}$ denotes the set of all natural numbers (i.e. positive integers).

Sometimes, we may wish to vary the index n over a different range. For instance, the sequence

$$a_0, a_1, a_2, \ldots, a_n, \ldots$$

may be written as $(a_n)_{n=0}^{\infty}$. If the range of n is implicitly understood from the context, we may also just denote the sequence by $(a_n)_n$. Sometimes, if the index n is understood from the context, we may also indicate the sequence by (a_n). (This is often done in the textbook. However, this should be done only when there is no chance that the reader might interpret it as the single element a_n.)

Example 1. Let us look at some examples:

(1) Consider the sequence $(a_n)_{n=1}^{\infty}$ where $n = (-1)^n$. If we list the individual entries, they are as follows:

$$-1, 1, -1, 1, \ldots$$

(2) The sequence $(1/n)_{n=1}^{\infty}$ has the following terms:

$$1, 1/2, 1/3, \ldots, 1/n, \ldots$$

(3) It is not necessary that all the terms should be distinct. For instance, we may have the sequence $(a_n)_{n=1}^{\infty}$ where $a_n = 1$ for all n.

$$1, 1, 1, \ldots, 1, 1, \ldots$$

(4) It is not even necessary that there should be a "nice" formula for the n-th term of the sequence. For example, we have the following sequence with no specific formula, and we are only listing the first few terms:

$$1, -1, 3/2, \pi, 42, \ldots$$

2. Why do we care about sequences?

Let us see a couple of examples to see why sequences are important. Sequences are ubiquitous in mathematics and you have been using them from your earliest school days. However, there are certain details which may not be spelt out in primary school.

2.1. Decimal expansions. For example, we know that the number 1/3 has the "decimal expansion" $0.333333\ldots$, where it is understood that the $3's$ continue "forever". In fact, we write the equality

$$1/3 = 0.3333\ldots. \tag{1}$$

What does the right hand side mean?

We know that 0.3 is just another notation for 3/10. Similarly, 0.33 is actually just a way of representing a certain sum:

$$0.33 = \frac{3}{10} + \frac{3}{100}$$

Similarly,

$$0.3333 = \frac{3}{10} + \frac{3}{100} + \frac{3}{1000} + \frac{3}{10000}.$$

So what does $0.3333\ldots$ mean? Does it mean the following "infinite" sum?

$$\frac{3}{10} + \frac{3}{100} + \frac{3}{1000} + \frac{3}{10000} + \ldots + \frac{3}{10^n} + \ldots$$

The problem here is that algebraically this kind of an infinite sum makes no sense. Addition is an operation that involves only two numbers. A sum involving three terms is interpreted as the number that is obtained by performing two addition operations, adding two numbers at a time.

$$x + y + z = (x + y) + z$$

In the same way, we can make sense of any *finite* sum. But an infinite sum does not make sense right away.

We need to *assign meaning* to infinite sums. This is a new concept that we need to create. One of the main objectives of this course is to make this precise.

So, let us see what we really mean by the equality in equation (1). It is mean to suggest that the numbers 0.3, 0.33, 0.333, etc. give increasingly accurate approximations of the number 1/3. None of them is equal to 1/3, but they come as close to 1/3 as desired. For degree of accuracy (whatever that means) we may desire, we can choose a sufficiently large n so that the n-th term in the sequence (in which the number of 3's after the decimal point is n) is good enough for our needs. This last sentence needs to be elaborated upon and made precise, which is something we will achieve in this course. We will then say that 1/3 is the "limit" of the sequence

$$0.3, 0.33, 0.333, 0.3333, \ldots.$$

2.2. Area of a circle. Another situation that subtly involves sequences occurs in the context of geometry. We know that the area of a circle of radius r is πr^2. But how does one make sense of "area" of a circle? The notion of area needs to be rigorously developed for this. We agree the area of a square of side a is a^2. We also expect that area of congruent regions should be equal and that the area of a union of disjoint regions should be equal to the sum of the area of the individual reasons. Using these properties, we can easily deduce that the standard formula for the area of an arbitrary triangle. Since any polygon can be decomposed into triangles, we can also make sense of the area of an arbitrary polygon.

However, this approach does not immediately tell us what the area of a circle should be. And yet, intuitively we know what we mean by the area of a circle. One of the most natural ways to make sense of it is as follows:

We pick n equidistant points on a circle. These form a regular polygon which we denote by P_n. We can then compute the area of the polygon P_n for each n. Then we consider the sequence $(Area(P_n))_{n=3}^{\infty}$. The area of the circle is then understood to be the "limit" of this sequence. However, one is then required to prove that this sequence has a limit. After all, not all sequences have limits! For example, the very first sequence given in Example 1 does not seem to "settle down" at all, and so does not have any limit.

3. About "numbers"

We saw above that the number $1/3$ can be seen as a "limit" of a certain sequence and thus denoted by the decimal expansion $0.333\ldots$ which has infinitely many terms. However, the number $1/3$ has a separate meaning which does not depend on being represented as a limit. Indeed, $1/3$ is the number which after multiplying by 3 gives the number 1. This leads us to a crucial question: Does such a number really exist?

This opens up a Pandora's box of uncomfortable questions. What is a "number"? What does it mean to say that a number "exists"? These questions require a serious and very long discussion. We will not attempt to answer this questions in this course. However, let us just have a quick, informal discussion.

The natural numbers are what we use for "counting". What this really means can be made precise in a set theory course, but let us just move along with the intuitive understanding we have of this notion. What are the negative numbers? We define negative integers as being those which satisfy certain algebraic equations. For example, -1 satisfies the equation $(-1) + 2 = 1$, etc. However, note that -1 is not really used for counting. It is only meant to be understood in terms of its algebraic properties.

Once this is done, we play a similar game and define the rational numbers as those that are solutions of some equations. For instance, $1/3$ is the number which satisfies the equation $(1/3) \times 3 = 1$. Again, note that we cannot really use the rational numbers for counting. There is no such thing as a collection that has $3/2$ objects in it.

There are two things to consider in such constructions. Firstly, it is not always a good idea to just randomly add new numbers to your number system by requiring them to satisfy certain algebraic properties. For example, suppose I want to add a number α to my number system such that it has the properties $\alpha + 1 = 4$ and $\alpha - 1 = 1$. This will create problems – do you see them?

The upshot is that while the above construction of rational numbers is easy to motivate, it needs to be verified carefully. It is not immediately clear that the rational numbers really exist. This can be established, but we will not do it in this course.

A second thing to note is that, again, rational numbers are no longer used for counting in the usual sense. However, we adjust our imagination and say that they can be used to denote "quantities", such as length. We start depicting rational numbers as points on the "number line". One of the points is marked as 0 and another point is marked as 1. The distance between them is defined to be 1 unit. For any given rational number x, we shift x units to the right from 0 and we marked the point reached by the symbol x. So, points that are to the left of 0 are marked by negative numbers.

It is important to note this change of intuition. We started by thinking of numbers as something that we use for counting, but we were forced to change our perspective and see them as representing quantities such as distance. The important thing to note here is that neither perspective is absolutely necessary. It does not matter what a number actually is. It is the properties it seems to have (algebraic or otherwise) that actually matter. In the long run, it is useful to internalize this way of looking at numbers.

4. Irrational numbers

Even when we start viewing numbers as something used to measure distances, we run into a problem.. We find that there are distances which cannot be measured by rational numbers. Indeed, the Pythagorus Theorem says that if x is the length of the hypotenuse of a right angled triangle in which the other two sides are both of unit length, then x^2 must equal 2. However, there is no rational number with this property.

Theorem 2. *There is no rational number x such that $x^2 = 2$.*

Proof. Suppose x is a rational number such that $x^2 = 2$. We write x as r/s where r and s are coprime integers. Then $r^2 = 2s^2$. Thus, 2 divides r, and so we may write $r = 2u$. Thus, $4u^2 = 2s^2$, i.e. $2u^2 = s^2$. But this implies that 2 divides s. Thus, 2 is a common divisor of both r and s, which contradicts our assumption that r and s are coprime. This contradiction shows that there is no rational number x such that $x^2 = 2$. $\square$

Thus, if we were to restrict ourselves to only working with the rational numbers, we will be left with some "holes" in the number line. Our notion of number should thus be expanded so that at least all the points on the line represent a number.

A warning – this entire discussion has been incredibly vague! We never really said what the number line is. We have adopted the standpoint that every point on the line is a number. We will, in fact, call these as "real numbers". But then the problem is that we do not yet have a *mathematical* definition of the line. We are unconsciously using our intuitive notion of a line which is inspired by our experience of the physical world.

There are multiple ways to make all this rigorous. We will not do that in this course. The purpose of this lecture was to simply point out that there are all these things that you may have been taking for granted.

5. Reading suggestions

The contents of this lecture correspond to Section 1.1 of the textbook.

The following reading suggestions are extra-curricular (hence, optional):

- Paul R. Halmos, *Naive Set Theory*, Undergraduate Texts in Mathematics, Springer, 1974.
  This text describes the construction of the natural numbers starting with the axioms of set theory.
- Section 8.6 of the textbook: This shows how the real numbers are constructed (assuming the existence of the rational numbers).